

Software Design Document: Smart Student Hub

Project Title: Smart Student Hub: Task Tracking & Deadline Reminder System for UNESWA Students

Name(s): Tamile Ntshalintshali

Nosipho Shabangu

Karabo Nkonyane

Lab Section: 04

Workstation: IDE LAB

Date: 13 May 2026

Contents

1. Introduction	2
1.1 Purpose	2
1.2 Scope	2
1.3 Overview	2
1.4 Reference Material	2
1.5 Definitions and Acronyms	2
2. SYSTEM OVERVIEW	2
3. SYSTEM ARCHITECTURE	3
3.1 Architectural Design	3
3.2 Decomposition Description	3
3.3 Design Rationale	4
4. DATA DESIGN	4
4.1 Data Description	4
4.2 Data Dictionary	4
5. COMPONENT DESIGN	5
6. HUMAN INTERFACE DESIGN	5
6.1 Overview of User Interface	5
6.2 Screen Images	6
6.3 Screen Objects and Actions	8
7. REQUIREMENTS MATRIX	9
8. APPENDICES	9

1. Introduction

1.1 Purpose

This Software Design Document (SDD) describes the architecture and system design of the Smart Student Hub. The document is intended for developers, lecturers, project supervisors and stakeholders involved in the development of the system.

1.2 Scope

The Smart Student Hub is a task tracking and deadline reminder system developed for UNESWA students. The system helps students manage assignments, tests, presentations, and academic schedules using a centralized platform. The system provides deadline alerts and reminders to reduce the chances of missing important academic activities. The main objective of the project is to improve academic organization, time management, and productivity among university students.

1.3 Overview

This document is organised into eight sections, covering the system architecture, data design, component algorithms, and human interface specifications required to build the Student Hub.

1.4 Reference Material

- IEEE Software Design Document Template
- Systems analysis and design notes
- Smart student hub Project proposal

1.5 Definitions and Acronyms

- SDD: Software Design Document
- LMS: Learning Management System
- SRS: Software Requirements Specification
- PDL: Procedural Description Language
- Reminder Notification – alert sent to students before deadlines

2. SYSTEM OVERVIEW

The Smart Student Hub is a web-based academic task management system developed for students at the University of Eswatini. The system allows students to organize and monitor academic activities such as assignments, tests, quizzes, and presentations. Students can create tasks, set deadlines, receive reminders, and manage their academic schedules from a single platform. The system aims to reduce stress caused by poor planning and missed deadlines.

Main users

- Student

Main features

- User registration and login
- Task Creation and Management
- Deadline Tracking
- Reminder notifications
- Dashboard overview

3. SYSTEM ARCHITECTURE

3.1 Architectural Design

The Smart Student Hub uses a three-tier architecture

1. Presentation layer

This layer contains the user interface where students interact with the system through forms, dashboards and notification

2 Application layer

This layer processes system operations such as:

- User authentication
- Task management
- Deadline reminders
- Notification handling

3 Database layer

Layer that stores:

- Student information
- Academic tasks
- Deadline schedules
- Notification

3.2 Decomposition Description

The system is decomposed into several key functional modules:

1. Authentication Module:

- User registration
- User login
- Passwords validation

2. Task Management Module:

- Add tasks
- Update tasks
- Delete tasks
- View task status

3. Deadline Reminder Module

- Monitor deadlines
- Generate reminders alerts
- Send notifications to users

4. Dashboard Modules:

- Displays upcoming deadlines
- Show pending tasks
- Display completed tasks

3.3 Design Rationale

The Client-Server model was chosen to ensure the system is accessible across different devices (laptops and smartphones) while maintaining a centralized database for data integrity. The Waterfall life cycle was selected because the system requirements and scope are stable and well-defined. The three-tier architecture was selected because it separate user interface, business logic and data storage. It also improves maintainability and support future system expansion. Lastly it ensures security and system organisation.

4. DATA DESIGN

4.1 Data Description

The information domain is transformed into a relational database structure. Entities such as "Students," "Assignments," and "Tests" are organized to reflect real-world academic relationships. The database supports user authentication, task management, deadline tracking and notification storage

4.2 Data Dictionary

Entity	Attributes	Description
Student	Student-ID, Name, Email, Passwords	Store student information
Task	Task-ID, Tittle, Description Deadline	Store academic tasks
Notification	Notification-ID, Message and date	stores reminder notification

5. COMPONENT DESIGN

User Login Function

Pseudocode

```
Begin
    INPUT username;
    INPUT password
    IF surname and password are correct
    THEN
        Display dashboard
    ELSE
        DIPLAY ERROR message
    END IF
END
```

Reminder Notification

```
Begin
CHECK task deadlines
IF deadline is near THEN
    SEND reminder notification
ENDIF
END
```

ADD TASK FUCTION

```
BEGIN
INPUT task tittle
INPUT description
INPUT deadline
SAVE task into database
Display success message
END
```

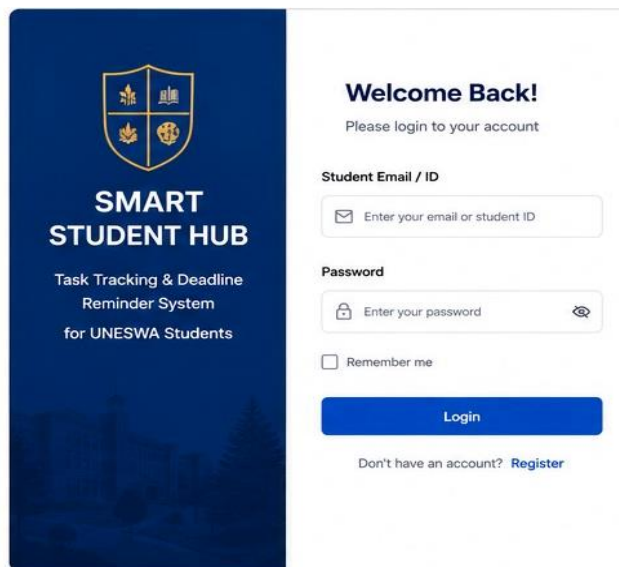
6. HUMAN INTERFACE DESIGN

6.1 Overview of User Interface

The Smart Student Hub interface is designed to be simple, user-friendly, and easy to navigate for UNESWA students. The system allows users to login into the system, create academic tasks, view deadlines, receive reminder alerts, monitor pending and completed tasks. The dashboard provides quick access to all important academic activities and reminders

6.2 Screen Images

Login Screen



The login screen features a dark blue sidebar on the left with the SMART STUDENT HUB logo and text: "SMART STUDENT HUB", "Task Tracking & Deadline Reminder System", and "for UNESWA Students". The main area is white and contains a "Welcome Back!" message, a prompt to login, and input fields for "Student Email / ID" and "Password". There is a "Remember me" checkbox and a "Login" button. A link to "Register" is provided for users without an account.

Welcome Back!
Please login to your account

Student Email / ID
Enter your email or student ID

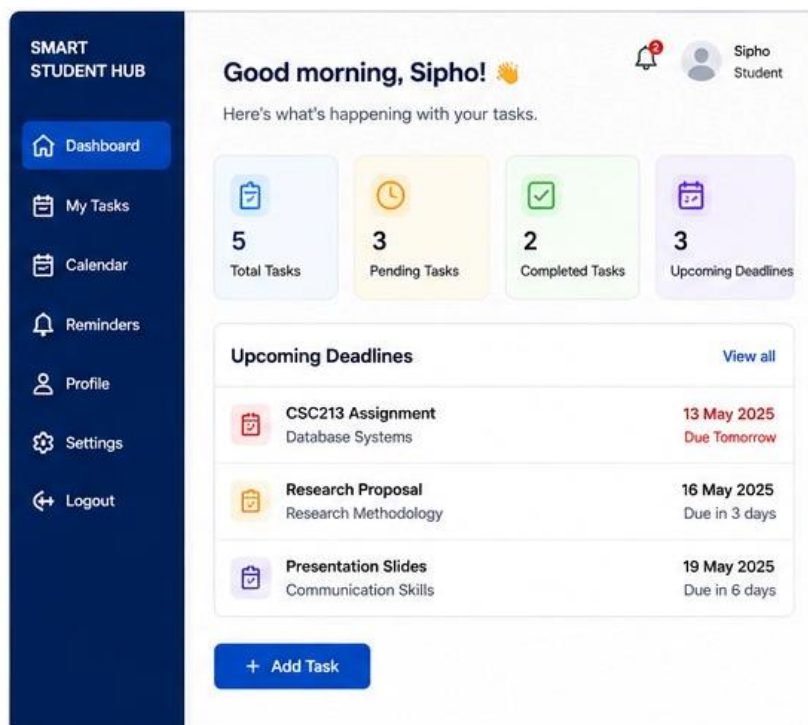
Password
Enter your password

☐ Remember me

Login

Don't have an account? [Register](#)

Dashboard Screen



The dashboard screen features a dark blue sidebar on the left with the SMART STUDENT HUB logo and a list of navigation items: Dashboard, My Tasks, Calendar, Reminders, Profile, Settings, and Logout. The main area is white and contains a "Good morning, Sipho!" greeting, a notification bell, a user profile icon, and a summary of tasks. Below this is a section for "Upcoming Deadlines" with a list of tasks and their due dates. A "View all" link is provided for the deadlines section. An "Add Task" button is located at the bottom.

SMART STUDENT HUB

Good morning, Sipho! 🌞

Here's what's happening with your tasks.

5 Total Tasks

3 Pending Tasks

2 Completed Tasks

3 Upcoming Deadlines

Upcoming Deadlines [View all](#)

	CSC213 Assignment Database Systems	13 May 2025 Due Tomorrow
	Research Proposal Research Methodology	16 May 2025 Due in 3 days
	Presentation Slides Communication Skills	19 May 2025 Due in 6 days

+ Add Task

Add tasks screen

← Add New Task

Task Title

Enter task title

Subject

Select subject

Description

Enter task description

Deadline Date

dd/mm/yyyy

Priority

Select priority

Cancel

Save Task

Task list screen

SMART STUDENT HUB

Dashboard

My Tasks

Calendar

Reminders

Profile

Settings

Logout

My Tasks

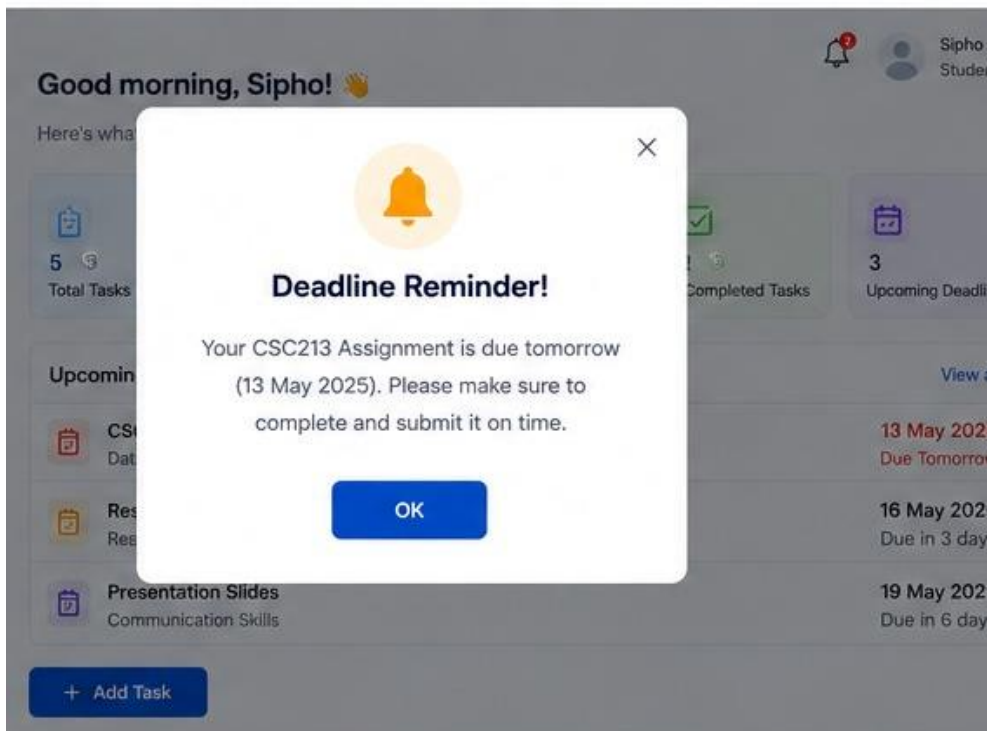
+ Add Task

All Pending Completed

Task Title	Subject	Deadline	Priority	Status	Actions
CSC213 Assignment	Database Systems	13 May 2025	High	Pending	 
Research Proposal	Research Methodology	16 May 2025	Medium	Pending	 
Presentation Slides	Communication Skills	19 May 2025	High	Pending	 
Quiz 2	Operating Systems	25 May 2025	Low	Completed	 
Lab Report	Software Engineering	28 May 2025	Medium	Completed	 

Showing 1 to 5 of 5 tasks

Reminder Notification Screen



6.3 Screen Objects and Actions

Screen Object	Action
Login Button	Authenticates user
Register Button	Creates new account
Add Task Button	Saves academic task
Edit Button	Updates task information
Delete Button	Removes task
Reminder Alert	Displays upcoming deadlines
Calendar Icons	Clickable dates to view specific tasks details

7. REQUIREMENTS MATRIX

Requirement ID	Authentication Module	Data Structure
FR-01	Allow student to register and log in	Authentication Module: handle user registration and secure login via the student's database table
FR-02	Allow student to add assignment	Assignment tracking module: provides forms to input titles and due dates into the assignments tables
FR-03	Allow student to add test date	Test management module: allow test, exam dates into to the test table
FR-04	Display deadline in calendar format	Calendar view interface: A front-end component that pull data from assignment and test table into a visual grid
FR-05	Send reminder before deadlines	Notification/Reminder System: A backend logic component that trigger alert based on the system.
FR-06	Update and delete task	Task Management: Interface: CRUD (Create, Read, Update, Delete) functionality within the dashboard

8. APPENDICES

Development Tools: PHP, MySQL, HTML, CSS, JavaScript

Database Schema: Detailed ER Diagram showing relationships.

Project Risks: Detailed mitigation strategies for internet and technical issues.

Training Schedule: Gantt chart for team skill development in SQL and GitHub.

Future improvement: SMS and email reminders, cloud backup system